

RailSentinel-24

Hardware v1.0 · Design targets

Control-power monitor and transient recorder for DC fast-charger cabinets

RailSentinel-24 (RS-24) mounts on the DIN rail inside a DC fast-charger cabinet and monitors the 24 V auxiliary supply that every relay driver, contactor coil and controller board depends on. It detects and records the inductive transients that reset controllers before they can write a log, timestamps them against a battery-backed clock, stores them locally, and reports them over Ethernet, WiFi or cellular — independently of the charger's own controller.

The measurement front-end sits on its own galvanically isolated domain, referenced to the rail; the ESP32 controller, storage and radios sit on the isolated secondary side. A phone tapped on the enclosure reads the last events over NFC without app, network or login.

12–62 V

DC rail range, one hardware variant for 12 / 24 / 48 V systems

< 2 μ s

hardware transient detection latency

 ± 0.15 V

absolute rail-voltage accuracy after calibration

5 kV_{RMS}

reinforced isolation, measurement to controller side

POWER INPUT

Supply source	from monitored rail	J_IN; no separate supply
Nominal input	24 V DC	12 V and 48 V systems supported
Operating input	9 – 62 V DC	80 V rated buck converter
Input protection	TVS + reverse polarity	SMBJ60A-class clamp; P-FET reverse protection, indefinite
Board power	< 0.5 W typ.	WiFi active; ~1 W peak on transmit
Hold-up through rail dips	planned	Supercapacitor hold-up scheduled for hardware v1.1

RAIL MEASUREMENT

Voltage range	0 – 62 V	20-bit digital power monitor, direct bus sensing
Voltage accuracy	± 0.15 V	absolute, after 3-point calibration
Voltage resolution	< 1 mV	20-bit conversion
Current range	± 3.2 A	full cabinet aux load, 50 m Ω Kelvin shunt
Trend sample rate	up to 860 SPS	rail voltage and current
Ripple trending	firmware	rolling baseline, PSU-degradation indicator

TRANSIENT CAPTURE

Detection	hardware comparator	< 2 μ s to interrupt, independent of firmware
Threshold	20 – 60 V adaptive	set from firmware, configurable per site
Peak amplitude	16-bit, ± 0.5 mV	analog peak-hold, digitised on the measurement side
Event envelope	50 ms window	0.3 – 1.2 ms/sample; sub- μ s waveform planned for v2.0
Timestamp	± 2 ppm RTC	battery-backed (CR2032), valid offline
Local storage	16 MB flash	$\geq 1,000$ -event ring buffer, survives power loss

ISOLATION AND SAFETY

Barrier	5 kV _{RMS} reinforced	digital isolator between measurement (rail-referenced) and controller domain
Isolated supply	push-pull transformer	5 kV _{RMS} rated
Service port	isolated UART	touch-accessible port double-insulated from mains-derived circuits
Function	monitoring only	RS-24 does not control, inhibit or switch any charger function

CONNECTIVITY

WiFi	802.11 b/g/n	main board; external antenna (U.FL)
Ethernet	10/100, RJ45	COM-ETH board
Cellular	NB-IoT / LTE-M	COM-LTE board (RS-24-F)
Protocol	MQTT, TLS	telemetry, events, configuration
NFC	ISO 14443 Type 2	passive tag, works unpowered; external antenna on IP67 variant
Field service	UART, OTA	signed firmware updates

ENVIRONMENTAL AND MECHANICAL

Operating temperature	–10 ... +70 °C	cabinet interior
Storage temperature	–20 ... +85 °C	
Humidity	5 – 95 % RH	non-condensing
Main board	65 × 55 mm	4-layer; M3 grid 58.5 × 48.5 mm
Expansion	FFC, 14-pin 0.5 mm	Sense-EXP and COM boards, removable
Vibration / shock	IEC 60068-2-6 / -27	DIN-rail equipment profile (design target)

SENSING EXPANSION (SENSE-EXP, OPTIONAL)

Temperature	4 × NTC, ± 0.5 °C	probe inputs, 1 kV fault-protected
Acoustic	MEMS microphone	5-band signature per event; contactor and fan health
Vibration	3-axis MEMS	fan / mechanical state

Correlation	vibration, acoustic, thermal	with Sense-EXP board, per event	Main bus	clip-on CT	AC input correlation
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VARIANTS

RS-24-S Standard DIN-rail enclosure, IP40. Ethernet or WiFi. For managed sites. NFC on the front face.	RS-24-F Field IP67 enclosure, cellular. External NFC antenna through the lid. For remote and motorway sites.	RS-24-M OEM module Bare main board on standoffs for factory integration. REST API, signed OTA. Per-unit licence.
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Status. Hardware v1.0 is in bench bring-up. All values are design targets from the hardware design documentation and are subject to change until validation is complete. The product is not yet CE marked and not commercially available; pilot units are supplied under a pilot agreement. Specifications will be replaced by measured values in Rev. 1.0 of this datasheet.

ORDERING AND CONFIGURATION

CONFIGURATION	BOARDS	ENCLOSURE	CONNECTIVITY	TYPICAL USE
RS-24-S Basic	Main	DIN rail, IP40	WiFi	Service companies, single sites
RS-24-S Standard	Main + COM-ETH	DIN rail, IP40	Ethernet	Managed CPO sites
RS-24-S Full	Main + Sense-EXP + COM-ETH	DIN rail, IP40	Ethernet	Root-cause analysis, thermal and acoustic correlation
RS-24-F	Main + COM-LTE (+ Sense-EXP)	IP67	NB-IoT / LTE-M	Remote sites without network
RS-24-M	Main (bare)	none	any	OEM factory integration

Enclosures: Italtronic Modulbox XTS (DIN) and Bopla Bocube (IP67). COM-LTE and RS-485 boards follow the main board and COM-ETH in the release sequence.

INSTALLATION SUMMARY

Mounting	35 mm DIN rail	6 M width; or M3 standoffs (RS-24-M)
Wiring	2 wires	J_IN from the 24 V rail; optional pass-through J_OUT for current measurement of the downstream load
Commissioning	NFC / UART	network credentials and thresholds set on first boot
Probe cables (Sense-EXP)	≤ 500 mm	300 V rated, 85 °C, flame retardant; route away from DC busbars
Service port	J2	isolated; for commissioning and field flashing only

DATA AND SOFTWARE

Telemetry	every 5 s (configurable)	rail voltage, current, temperatures, vibration, acoustic level, uptime
Events	on trigger	peak amplitude, duration, envelope, timestamp, correlated sensor snapshot
Dashboard	web	per-device trends, event log, fleet view; commissioning report (planned)
Integration	MQTT, REST API	OEM data interface for RS-24-M
Security	TLS, per-device credentials, signed OTA	designed to Cyber Resilience Act requirements

COMPLIANCE ROADMAP

CE (LVD, EMC, RED, RoHS)	planned	pre-compliance testing scheduled; certification targeted 2027
Isolation	IEC 62368-1	reinforced barrier and isolated service port designed to §5.4 double-insulation requirement
Cyber Resilience Act	designed for	secure development, update capability, vulnerability handling

REVISION HISTORY

Rev. P1	2026-09	First preliminary release based on hardware v1.0 design documentation. All values are design targets.
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